

**PhD Analysis Examination**  
**May 2016**

Do THREE problems from Part A and THREE problems from Part B. Answer each question on a separate sheet of paper. Write solutions in a neat and logical fashion, giving complete reasons for all steps.

**Part A**

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1. Consider Lebesgue measure on the real line  $\mathbb{R}$ . Give an example for each of the following, if possible. If not possible, give a brief explanation.

A sequence  $(f_n)$  in  $L^1(\mathbb{R})$  converging to an  $f$  in  $L^1(\mathbb{R})$ ...

- a) ...in the  $L^1$  norm but not in measure,
  - b) ...in measure but not in the  $L^1$  norm,
  - c) ...in the  $L^1$  norm but not a.e.,
  - d) ...a.e. but not in measure.
2. Let  $\nu$  be a signed measure on  $(X, \mathcal{M})$ .
- a) Define what it means for a set to be *positive*, *negative* or *null* for  $\nu$ , and state the Hahn decomposition theorem.
  - b) Prove there exist unique positive measures  $\nu^+, \nu^-$  on  $\mathcal{M}$  so that  $\nu = \nu^+ - \nu^-$  and  $\nu^+ \perp \nu^-$ .
3. Let  $\mu$  be a finite, regular Borel measure on  $[0, 1]$ . Suppose that

$$\int_0^1 x^n d\mu = 0 \quad \text{for } n = 0, 1, 2, \dots$$

Prove that  $\mu = 0$ .

4. Let  $(X, \mathcal{M}, \mu)$  be a  $\sigma$ -finite measure space. Let  $\mathcal{N}$  be a sub- $\sigma$  algebra of  $\mathcal{M}$  and let  $\nu$  be the restriction of  $\mu$  to  $\mathcal{N}$ . Prove that for each  $f \in L^1(\mu)$ , there exists a  $g \in L^1(\nu)$  such that

$$\int_E f d\mu = \int_E g d\nu$$

for all  $E \in \mathcal{N}$ .

## Part B

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1. Let  $\mathcal{X}$  be a normed vector space. Say a sequence  $(x_n)$  from  $\mathcal{X}$  converges *weakly* to  $x \in \mathcal{X}$  if  $f(x_n) \rightarrow f(x)$  for all  $f \in \mathcal{X}^*$ . Prove that if  $\mathcal{M}$  is a norm closed subspace of  $\mathcal{X}$ , and  $(x_n)$  is a sequence in  $\mathcal{M}$  converging weakly to  $x \in \mathcal{X}$ , then  $x \in \mathcal{M}$ .
2. Let  $\mathcal{H}$  be a Hilbert space and  $\mathcal{M}, \mathcal{N}$  closed subspaces of  $\mathcal{H}$ . Prove that if  $\mathcal{M} \perp \mathcal{N}$ , then  $\mathcal{M} + \mathcal{N}$  is closed.
3. Fix  $1 < p < \infty$ , let  $\mu$  be a  $\sigma$ -finite measure, and let  $(f_n)$  be a sequence in  $L^p(\mu)$ . Suppose that there exists a function  $f \in L^p(\mu)$  such that

$$\int f_n g d\mu \rightarrow \int f g d\mu$$

for every  $g \in L^q(\mu)$ , where  $\frac{1}{p} + \frac{1}{q} = 1$ .

- a) Prove that  $\sup_n \|f_n\|_p < +\infty$ .
  - b) Prove that  $\|f\|_p \leq \limsup \|f_n\|_p$ .
4. Let  $(\varphi_n)$  be a sequence in  $L^1(\mathbb{R})$  with the following properties:
    - i)  $\varphi_n \geq 0$  for all  $n$ ,
    - ii)  $\int_{\mathbb{R}} \varphi_n(x) dx = 1$  for all  $n$ , and
    - iii) for every  $\delta > 0$ ,  $\lim_{n \rightarrow \infty} \int_{|x| > \delta} \varphi_n(x) dx = 0$ .

Prove that  $f * \varphi_n \rightarrow f$  in  $L^1(\mathbb{R})$ .